

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda (UTC+3)

PROFESSIONAL SUMMARY

Full-stack technical lead (TypeScript/Node.js) based in Africa, with 6+ years building and running production systems. Led the end-to-end, zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP reporting directly to the CTO — owning the architecture, the sequencing, and the highest-risk implementations myself — while cutting \$5,000/month in infrastructure cost. Deep in event-driven architecture from production scars: built an idempotent, real-time two-way MongoDB↔Firestore sync on Google Pub/Sub that carried parallel live traffic for months with zero data loss. Architected event-driven microservices (Apache Kafka, Kubernetes) adopted across web, mobile, and USSD clients, and led a team of 4 engineers. Experience with fintech-adjacent money paths: DeFi/Web3 backends and revenue-critical e-commerce features. Now applying the same architecture-plus-hands-on leadership to real-money commercial systems.

CORE COMPETENCIES

TypeScript / Node.js

Event-Driven Architecture (Pub/Sub, Kafka)

Idempotent Message Processing

Serverless Backends (Cloud Functions)

API Design & Versioning (REST, gRPC)

Zero-Downtime Migrations

Kubernetes & Docker

Technical Leadership & Mentorship

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer (technical anchor, migration) — US Remote

- Led the end-to-end migration of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to the CTO — owned service boundaries, data models, cutover sequencing, and rollback strategy
- Built a real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub — idempotent message processing with conflict handling (the same dead-letter/redelivery discipline as SQS/EventBridge architectures)
- Implemented Express Shipping end to end, generating an additional \$40 revenue per order — a revenue-critical money path from backend pricing to checkout
- Built a 3D visualisation feature for customers using video overlay and ffmpeg, increasing buyer conversion
- Saved the company \$5,000/month by migrating all services off AWS; migrated file storage from Amazon S3 to GCS via a Python script

MTailor (Contract) Mar 2022 – Jul 2022

Backend Engineer, Contractor — US Remote

- Hired on an unfamiliar Firebase/GCP stack; built the MongoDB→Firestore migration scripts with complex processing logic and ramped fast enough to write the SDK documentation (Node.js and Python) other engineers onboarded from
- Trained the Ops team on the new Firebase dashboard and backup procedures — converted to full-time and migration leadership within months

Mind2matter Jan 2022 – Jul 2022

Software Engineer (Full-stack) — US Remote

- Built backends for DeFi applications using Web3 and Node.js for a US agency with demanding clients, reporting directly to the CTO
- Delivered production React UIs from Figma designs under tight timelines — componentized design-system approach

Dr Wealth Aug 2021 – Jan 2022

Software Engineer (Full-stack) — Singapore Remote

- Led optimisation of a PWA delivering real-time stock market prices: rates fetched from Morningstar APIs, cached, and served current across 2M+ Firestore records
- Extended backend REST APIs with Firebase Cloud Functions and Express.js; shipped responsive UI pages in React and Tailwind

CodeBits Jan 2020 – Jul 2021

Software Engineer & Team Lead — Uganda

- Led a team of 4 developers building legal tech systems for NGOs in Uganda — set the technical direction and reviewed the team's designs and code
- Architected an event-driven microservices backend for FIDA Uganda's case management system using Apache Kafka, Docker, and Kubernetes — architecture adopted across web, mobile, and USSD clients

- Built a USSD service for legal aid providers with a **versioned gRPC/protobuf contract** multiple third-party providers integrated against with no bespoke per-provider code — multi-provider, emerging-market infrastructure
- Developed a cross-platform mobile app for LASPNET using React Native and Expo with Firebase push notifications

PROJECTS

Real-time Pub/Sub Sync Pipeline

Live MongoDB ↔ Firestore two-way sync supporting parallel production traffic during a months-long migration — idempotent processing, conflict handling, zero data loss.

Node.js, Google Pub/Sub, MongoDB, Firestore

Healthcare WhatsApp Chatbot (AI-backed)

Integrated OpenAI + Google Vision for prescription OCR via the WhatsApp Business API — AI features shipped to production.

Node.js, OpenAI API, Google Vision, WhatsApp Business API

Marketplace Recommendation Engine

Collaborative + content-based filtering engine in MongoDB for service-provider matching in a live marketplace.

Node.js, MongoDB

EDUCATION

BSc Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, TypeScript/JavaScript, Firebase, GCP (Pub/Sub, Cloud Functions, Firestore), React, MongoDB, SQL, Python, Linux, Git

Intermediate: AWS (S3, EC2, EBS), Go, Rust, React Native **Languages:** English, Kiswahili